Rifat Rafiu

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Professional Summary

Final-semester Materials Science and Engineering student at KUET (expected August 2026) with hands-on experience in materials characterization, nanoparticle synthesis, mechanical testing, and DFT-based computational modeling. Completed industrial training at BSRM Steel, gaining practical exposure to steelmaking, continuous casting, and quality assurance. Skilled in microstructural analysis, technical reporting, and data-driven problem solving, with a strong foundation in metals, polymers, and composites. Eager to contribute to industrial R&D, product development, and quality engineering.

Academic credentials

Bachelor of Science in Materials Science and Engineering ***March 2022–Present***
Khulna University of Engineering & Technology (KUET)
CGPA: **3.55**/4.00 (up to 4th Year, 1st Semester)

Industrial Experience

BSRM Steels Ltd., Chattogram
Industrial Trainee - Production (SMS & CCM Department)
February 2025 - March 2025

- Observed steelmaking operations in the Steel Melting Shop (SMS).
- Studied Continuous Casting Machine (CCM) production processes and workflow.
- Learned industrial quality control, production monitoring, and process management practices.
- Gained exposure to workplace safety, manufacturing operations, and production planning.
- Worked closely with engineers to understand operational challenges and efficiency improvement opportunities.

Research Experience

Undergraduate Research Work ***September 2025– Present***
Khulna University of Engineering & Technology (KUET)
Photocatalytic Performance of Cobalt Oxide Nanostructures: A Combined Experimental-DFT Investigation

- Synthesized and characterized Co_3O_4 nanostructures using a co-precipitation method for photocatalytic applications.
- Investigated photocatalytic degradation performance by correlating optical band gap, charge transport, and surface properties.
- Applied DFT calculations alongside experiments to analyze structural and charge morphology, optical, and photocatalytic activity enhancement.

Publications

1. Mahmud, M. N., **Rafiu, R**., Paul, S. K., Enam, S. E., & Hasan, M. M. (2026, January). Hydrothermally Synthesized Urea-Derived Co3O4 Nanoparticles for Efficient Photocatalytic Degradation of Organic Dye.

In *2026 5th International Conference on Electrical, Computer & Telecommunication Engineering (ICECTE)* (pp. 1-6). IEEE. https://doi.org/10.1109/ICECTE69292.2026.11429372

2. **Rafiu, R.,** Hasan Saikot, M. S., Apon, I. A., Boukhris, I., El-Rayyes, A., Khan, M. T., ... & Rahman, M. A. (2025). Comprehensive DFT and SCAPS-1D Study of Structural, Electronic, Optical, Mechanical, Phonon, Thermoelectric, and Photovoltaic Properties of Lead-Free Z3BrO (Z= K, Rb, Cs, and Fr) Anti-Perovskites. Journal of Computational Chemistry, 46(24), https://doi.org/10.1002/jcc.70221
3. **Rafiu, R**., Kriaa, K., Saikot, M. S. H., Rahman, M. A., Apon, I. A., Sfina, N., ... & Talukder, I. A. (2025). Comprehensive first-principles analysis and device simulations of vacancy-ordered D2CeX6 double perovskites for high-efficiency lead-free solar cells. *RSC advances*, *15*(52), 44711-44748. https://doi.org/10.1039/d5ra07162c
4. **Rafiu, R**., Kriaa, K., Rahman, M. A., Apon, I. A., AlFaify, S., Maatki, C., ... & Elboughdiri, N. (2026). Unveiling the stability and optoelectronic potential of alkali-based L3FO anti-perovskites: a first-principles perspective. The European Physical Journal Plus, 141(1), 32. https://doi.org/10.1140/epjp/s13360-025-07239-3 .

Leadership and activity

General Member
Renewable Energy & Nanoelectronics Research Lab (RENRLab)| March 2025 - Present

- Perform computational investigation of materials' properties.
- Analyze and interpret scientific research data.
- Collaborate on peer-reviewed research publications.

Member, Cultural Committee
Amar Ekushey Hall, KUET | Oct 2024 - Oct 2025

- Coordinated student programs and cultural events.
- Worked in multidisciplinary teams to plan and execute activities.
- Developed organizational, communication, and teamwork skills.

Q&A Instructor
Udvash Academic & Admission Care | Mar 2023 - Aug 2025

- Conducted academic support sessions for students.
- Improved communication, mentoring, and problem-solving abilities.
- Guided students in developing analytical approaches to academic challenges.

Technical Skills

- **Data Analysis:** MS Excel, OriginPro, Data Visualization, Statistical Analysis
- **Engineering & Manufacturing:** Process Monitoring, Quality Assessment, Metallurgical Testing, Materials Characterization
- **Software:** Python, AutoCAD, MS Word, MS PowerPoint
- **Laboratory & Industrial Tools:** XRD, FTIR, SEM, Universal Testing Machine, Hardness Testing, Furnace Operation, Manufacturing Shop Equipment

Achievements

- Presented research at the IEEE-supported ICECTE 2026 International Conference.
- Recipient of Government Scholarships in JSC, SSC, and HSC examinations for academic excellence.